ACTIVITY 1.1

The H–R diagram—page 1



Skills: interpreting, numeracy

The temperature of a star can be estimated from its colour. The temperatures of several stars have been placed on a scale from 0 to 70 for ease of graphing. We will call these units 'SF units'. In reality, these values represent temperatures of several thousand degrees.

Apparent brightness is a measure of the brightness of stars as they appear to us on Earth. Just because a star appears brighter than another when viewed from Earth, it doesn't mean it is actually brighter, as its brightness may be due to it being much closer to us. To compare how bright stars actually are, we need to compare their brightness if viewed from the same distance. This may be done theoretically, and we then use the term 'absolute brightness'. The lower the value, the brighter the star. When the scale was first used, the brightest star was given absolute magnitude zero, but since then, brighter stars have been discovered and have had to be assigned negative values.

In 1913, Danish astronomer Ejnar Hertzsprung and American astronomer Henry Russell independently developed a way of displaying temperature and brightness data for stars; this became known as the Hertzsprung–Russell diagram, or H–R diagram for short. Use the data below for several stars to construct your own H–R diagram on the grid provided on page 2 of this activity. Our Sun's values are 41.5 and 4.6. Indicate the position of the Sun on your H–R diagram. 'Spectral class' is the temperature in SF units.

Spectral class	Absolute magnitude
21	–0.4
51	6
20	11
19	0
41.5	4.6
20	11.5
50	–1
52.1	2.3
20	13
67.5	14
62.5	–5.9
62	9.6
13	–3
65	13.4
11	–5
65	12.1
40	4.5
50.1	0.5
12.7	–1.8
66	13.2
9.9	–6.2
67	–1.1
65.5	15.1
50	6
9.3	–3.7
41	–3.7
55	7
12	–3.3
12.1	–3.7
19.8	0
57	8.3
64.4	12.7

Spectral class	Absolute magnitude
21	–0.1
65	13.2
50	1
64.2	12.5
51.1	–0.4
61	9
22.7	1.3
10	–4.7
10	–3.9
37.5	–4.6
42	4
49.9	–0.7
13	–7.5
64	10.2
54	–0.4
12	–2.6
54	–0.6
61	9.3
62	–0.8
55	–1
62	10
63	–2.6
11	–3.7
67	15
64.1	12.2
64	10
65.9	15.8
11	–3.4
50	0.1
59.9	0
62	10.2
19.9	–0.5

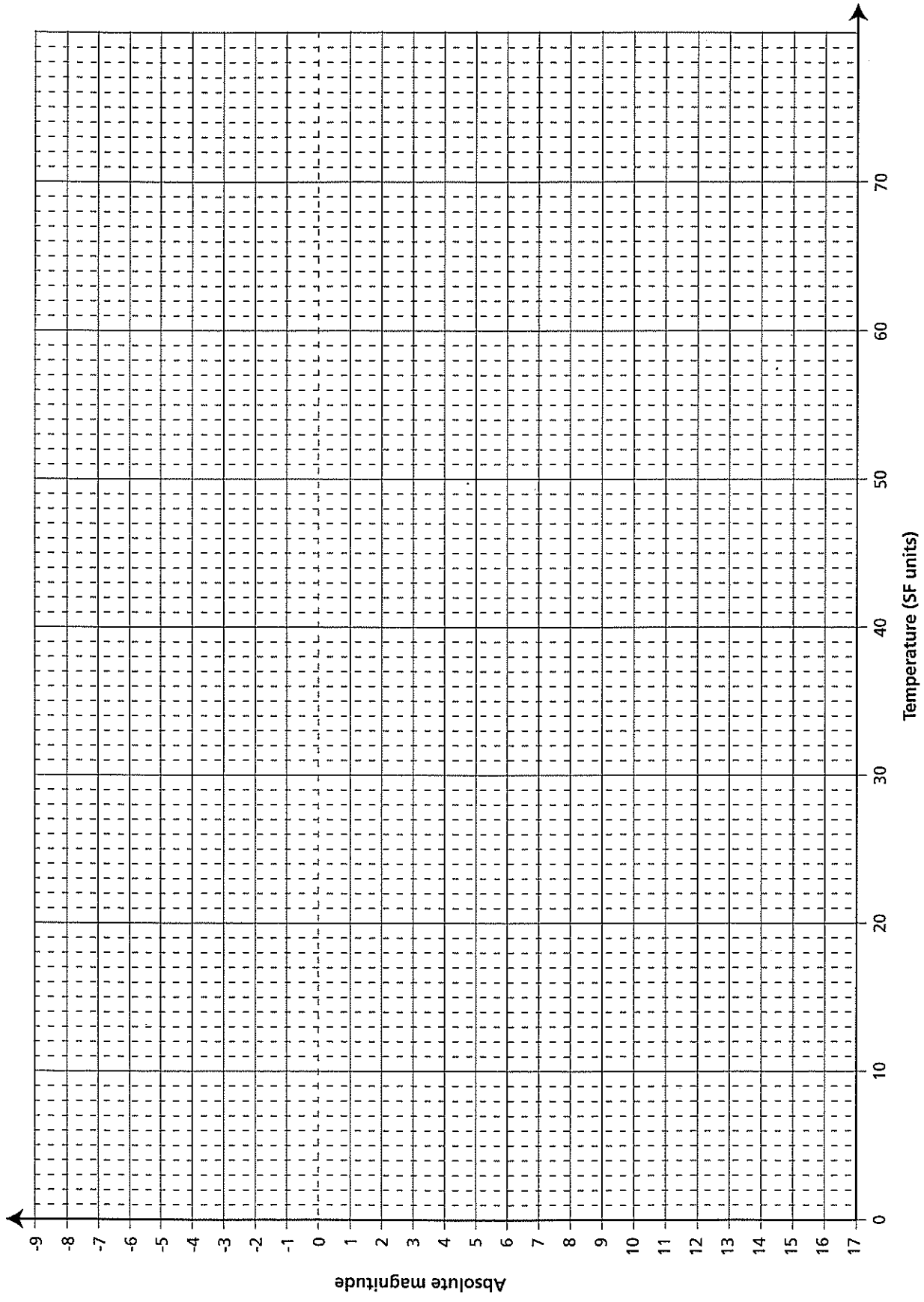
Spectral class	Absolute magnitude
21	–7.3
64	10.4
65	12.3
9.9	–6.6
52.1	0.5
21	0.4
9.3	–3.8
26.3	2.2
29.2	–4.8
65	12.5
19.8	–0.8
64	11
61	10.3
64.1	12.3
52	–0.4
65.5	14.9
19.9	–0.2
64	11.4
26.2	1.4
17.5	0.4
55.1	–0.7
63	11
15	–7
50	15.7
19	–1
52	0
60	9
30	12.9
60	8.8
34.8	–4.4
60	10.8
10.5	–3.8

Spectral class	Absolute magnitude
65	11.9
62	–1.5
28	2.1
29.1	–4.4
51.7	6
51	0.5
67	15.1
54.4	–4.5
22.5	–2
66	12.7
60	8.6
50.4	6.5
55	7.7
45	5.3
50	–0.5
31	1.4
65	15.4
65	15
45.9	4.6
49.9	–1.2
23.4	0.5
64.2	13.1
19	–1.8
65	12.7
65	12
65	11.8
12	–3
9.9	–4.2
57	8.5
35	2.5
64	11.1
0.5	–7

Spectral class	Absolute magnitude
16	–2.1
10.5	–5
54	–0.8
63.5	12
17	–2.1
67	13
37.8	–4.8
55	7.5
64	11.2
48	5.8
62.1	9.1
16.5	–0.9
14	–2.3
40	14
17	–7
10	–6.8
67	14.6
65	10.7
24.3	0.3
21.7	0.6
20.3	1.3
62	10.5
5.8	–7
49	6
20.9	0.3
29.9	–4.5
10.5	–4.5
15	0
55	5.7

The H–R diagram—page 2

Name:



The H–R diagram—page 3

You will need to refer to the graph you constructed.

1 Most of the points on an H–R diagram lie in a region called the 'main sequence'. Label this on your graph.

2 Stars in the region of the points plotted near (F5, –4.5) are called **supergiants**. Label six or more supergiants on your graph.

3 Stars in the region of the points plotted near (K3, 0) are called **giants**. Label 10 or more giants on your graph.

4 Stars in the region of the points plotted near (A0, 11.5) are called **white dwarfs**. Label three or more white dwarfs on your graph.

5 Star colours are divided into seven spectral classes: O (hottest class), B, A, F, G, K and M (coolest class). Each of these classes may be further divided into 10 subcategories numbered from 0 to 9, with zero denoting the hottest stars in a category. The temperature values used in your graph correspond to the spectral classes as shown below. Mark the spectral classes on your graph. You may also wish to colour your H–R graph as indicated by the last column of the table.

Spectral class	Horizontal axis range	Colour
O	0 to 10	Blue
B	10 to 20	Light blue
A	20 to 30	Light blue (left) fading to white (right)
F	30 to 40	White (left) changing to yellow (right)
G	40 to 50	Yellow (left) changing to orange (right)
K	50 to 60	Orange (left) changing to red (right)
M	60 to 70	Red

6 Include the following information on your H–R graph.

Spectral class value	Temperature
Right end of class O	20 500°C
Right end of class B	9 430°C
Right end of class A	6 930°C
Right end of class F	5 700°C
Right end of class G	4 400°C
Right end of class K	3 040°C

7 Indicate the direction on your graph to move for:

a smaller stars

b cooler stars

c smaller and cooler stars.